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Bansilal Ramnath Agarwal Charitable Trust's
VISHWAKARMA INSTITUTE OF TECHNOLOGY, PUNE – 411037.
 (An Autonomous Institute Affiliated to Savitribai Phule Pune University)

Examination: ESE

Year: 2024-25

Branch: DOME-CS/ENTC/IC/AI/IT/CSAI

Subject: Database Management Systems

Subject Code: CS2227

Max. Marks: 60

Total Pages of Question Paper: 02

Day & Date: 13/11/2024

Time: 11:00AM TO 1:00PM

Instructions to Candidate

1. All questions are compulsory.
2. Neat diagrams must be drawn wherever necessary.
3. Figures to the right indicate full marks.

Q.N.	CO No	BT* No		Max marks																														
Q.1.	1	1,2	A] Define DBMS. Explain any four CODD's rule.	5																														
	1	3	B] Draw an ER Diagram for a University Database with the following entities: Student, Department, Course, and Faculty. Ensure the diagram includes all entity types, attributes, relationships, cardinalities, and the constraints listed below: • A student has a unique ID and can enroll in multiple courses. • Each faculty member belongs to a department and can teach multiple courses. • Every student receives a grade for each enrolled course. • Courses must be associated with at least one department. Note: Students can make reasonable assumptions to elaborate on the problem if needed.	5																														
Q.2.	2	3	A] Consider the Employee entity with the following attributes: EmployeeID, Name, Email, PhoneNumber, SSN, DepartmentID, and ManagerID. Explain the distinctions among the following terms: • Primary Key • Foreign Key • Candidate Key • Super Key	4																														
	2	4	B] What is meant by Normalization ? Explain the criteria for BCNF (Boyce-Codd Normal Form) and analyze whether the given relation is in BCNF . <table border="1"> <thead> <tr> <th>STUDENT_ID</th><th>STUDENT_NAME</th><th>COURSE_ID</th><th>COURSE_NAME</th><th>INSTRUCTOR</th></tr> </thead> <tbody> <tr> <td>101</td><td>Alice</td><td>C001</td><td>Data Science</td><td>Dr. Smith</td></tr> <tr> <td>102</td><td>Bob</td><td>C002</td><td>Machine Learning</td><td>Dr. Miller</td></tr> <tr> <td>103</td><td>Carol</td><td>C001</td><td>Data Science</td><td>Dr. Smith</td></tr> <tr> <td>104</td><td>Dave</td><td>C003</td><td>Cloud Computing</td><td>Dr. Johnson</td></tr> <tr> <td>105</td><td>Eve</td><td>C002</td><td>Machine Learning</td><td>Dr. Miller</td></tr> </tbody> </table>	STUDENT_ID	STUDENT_NAME	COURSE_ID	COURSE_NAME	INSTRUCTOR	101	Alice	C001	Data Science	Dr. Smith	102	Bob	C002	Machine Learning	Dr. Miller	103	Carol	C001	Data Science	Dr. Smith	104	Dave	C003	Cloud Computing	Dr. Johnson	105	Eve	C002	Machine Learning	Dr. Miller	6
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			OR																															
	2	4	B] Analyze and justify whether the following relation is in 2NF. R(WXYZ) where the functional dependencies are given as $\{WX \rightarrow YZ, Y \rightarrow W, Z \rightarrow X\}$	6																														
Q.3.	3	3,5	A] Consider the following schema: employee (emp_id, emp_name, department, salary) project (proj_id, proj_name, department, budget)																															

			assignment (emp_id, proj_id, hours_worked) 1. Find the names of all departments in the employee relations, and remove duplicates. 2. Find the employee name, project name, and hours worked for employees assigned to the "Alpha" project. 3. Find the average salary for each department.	6																																												
	3	2	B] By considering suitable examples, Explain DML Queries.	4																																												
Q.4.	5	1,2	A] i] What is Transaction? Explain Commit, rollback and Save point commands. ii] Explain indexing concepts related to database storage.	4 2																																												
			OR																																													
	5	1,4	A] What do you mean by serializability? Analyze following schedule for conflict serializability and convert if it not. <table><tr><th>Time</th><th>T1</th><th>T2</th><th>T3</th></tr><tr><td>t1</td><td>Read(X)</td><td></td><td></td></tr><tr><td>t2</td><td></td><td></td><td>Read(Y)</td></tr><tr><td>t3</td><td></td><td></td><td>Read(X)</td></tr><tr><td>t4</td><td></td><td>Read(Y)</td><td></td></tr><tr><td>t5</td><td></td><td>Read(Z)</td><td></td></tr><tr><td>t6</td><td></td><td></td><td>Write(Y)</td></tr><tr><td>t7</td><td></td><td>Write(Z)</td><td></td></tr><tr><td>t8</td><td>Read(Z)</td><td></td><td></td></tr><tr><td>t9</td><td>Write(X)</td><td></td><td></td></tr><tr><td>t10</td><td>Write(Z)</td><td></td><td></td></tr></table>	Time	T1	T2	T3	t1	Read(X)			t2			Read(Y)	t3			Read(X)	t4		Read(Y)		t5		Read(Z)		t6			Write(Y)	t7		Write(Z)		t8	Read(Z)			t9	Write(X)			t10	Write(Z)			6
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	5	2	B] State and explain the ACID Properties.	4																																												
Q.5.	4	1	A] Define Inter query and Intra query parallelism with example.	4																																												
	4	2	B] Explain different types of Distributed Database management system.	4																																												
	4	2	C] Explain terms Speed up and Scale up.	2																																												
Q.6.	6	2	A] Explain the term Big Data.	3																																												
	6	1	B] What are NOSQL Databases? Explain their key features .	4																																												
	6	2	C] Explain OLAP Vs. OLTP.	3																																												

CO Statements:

CO1: Design data models as per data requirements of an organization
CO2: Synthesize a relational data model up to a suitable normal form
CO3: Develop a database system using relational queries and PL/SQL objects
CO4: Apply indexing techniques and query optimization strategies
CO5: Understand importance of concurrency control and recovery techniques
CO6: Adapt to emerging trends considering societal requirements

***Blooms Taxonomy (BT) Level No:**

1. Remembering; 2. Understanding; 3. Applying; 4. Analyzing; 5. Evaluating; 6. Creating